Description

Image





Caption

1. Slippers. © Zhangzhou Yongxin Trade Co. 2. Running shoes. © Adidas

The material

Ethylene-Vinyl-Acetate elastomers (EVA) are built around polyethylene. They are soft, flexible and tough, and retain these properties down to -60 C. Fillers improve both hardness and stiffness, but with some degradation of other properties. EVAs blend well with PE because of their chemical similarity. EVA is available in pastel or deep hues; it has good clarity and gloss. It has good barrier properties, little or no odor, is UV resistance and FDA-approval for direct food contact. The toughness and flexibility is retained even at low temperatures and it has good stress-crack resistance and good chemical resistance. EVA can be processed by most normal thermoplastic processes: co-extrusion for films, blow molding, rotational molding, injection molding and transfer molding.

Compositional summary

(CH2)n-(CH2-CHR)m

General properties

Density		945	-	955	kg/m^3
Price	*	0.944	-	1.02	GBP/kg
Date first used		1972			

Mechanical properties

Young's modulus		0.01	-	0.04	GPa
Shear modulus		0.008	-	0.01	GPa
Bulk modulus	*	1.3	-	1.4	GPa
Poisson's ratio	*	0.47	-	0.49	
Yield strength (elastic limit)		12	-	18	MPa
Tensile strength		16	-	20	MPa
Compressive strength		13.2	-	19.8	MPa

Elongation		730	-	770	% strain
Fatigue strength at 10^7 cycles	*	12	-	12.8	MPa
Fracture toughness	*	0.5	-	0.7	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.34	-	0.83	

Thermal properties

Melting point	*	145	-	185	°C
Glass temperature	*	-73.2	-	-23.2	°C
Maximum service temperature		46.9	-	51.9	°C
Minimum service temperature	*	-123	-	-73.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.3	-	0.4	W/m.°C
Specific heat capacity	*	2e3	-	2.2e3	J/kg.°C
Thermal expansion coefficient		160	-	190	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	3.16e21	-	1e22	µohm.cm
Dielectric constant (relative permittivity)		2.9	-	2.95	
Dissipation factor (dielectric loss tangent)		0.005	-	0.022	
Dielectric strength (dielectric breakdown)		26.5	-	27	MV/m

Optical properties

Transparency		Opaque			
Refractive index		1.48	-	1.49	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		3	-	4	
Moldability		4	-	5	
Machinability		3			
Weldability		2			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Unacceptable			
Soils, alkaline (clay)		Excellent			

Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Limited use
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Unacceptable
Chloroform	Unacceptable
Crude oil	Unacceptable
Diesel oil	Unacceptable
Lubricating oil	Unacceptable
Paraffin oil (kerosene)	Unacceptable
Petrol (gasoline)	Unacceptable
Silicone fluids	Excellent
Toluene	Unacceptable
Turpentine	Unacceptable
Vegetable oils (general)	Acceptable
White spirit	Unacceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Limited use

Ethyl alcohol (ethanol)	Unacceptable
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Unacceptable

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Poor

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 3.06	-	3.37	kg/kg
Embodied energy (virgin grade production)	* 85.5	-	94.3	MJ/kg
Water usage	* 2.66	-	2.94	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.2	-	9.2	MJ/kg
Polymer molding (injection, blow), embodied energy	* 24.9	-	31.9	MJ/kg
Machining, coarse, embodied energy	* 15.8	-	18.1	MJ/kg
Machining, fine, embodied energy	* 18.3	-	22.2	MJ/kg
Machining, grinding, embodied energy	* 20.8	-	26.2	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.542	-	0.611	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.78	-	2.33	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.04	-	1.19	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.2	-	1.45	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.36	-	1.72	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.13	-	1.25	kg/kg
Embodied energy, recycling	* 13.7	-	15.1	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 39.2	-	41.2	MJ/kg
Combustion CO2	* 2.82	-	2.97	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Recycle mark



Supporting information

Design guidelines

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Typical uses

Medical tubes, milk packaging, beer dispensing equipment, bags, shrink film, deep freeze bags, co-extruded and laminated film, closures, ice trays, gaskets, gloves, cable insulation, inflatable parts, running shoes.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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